

Assignments & Tutorial

- **Assignments**

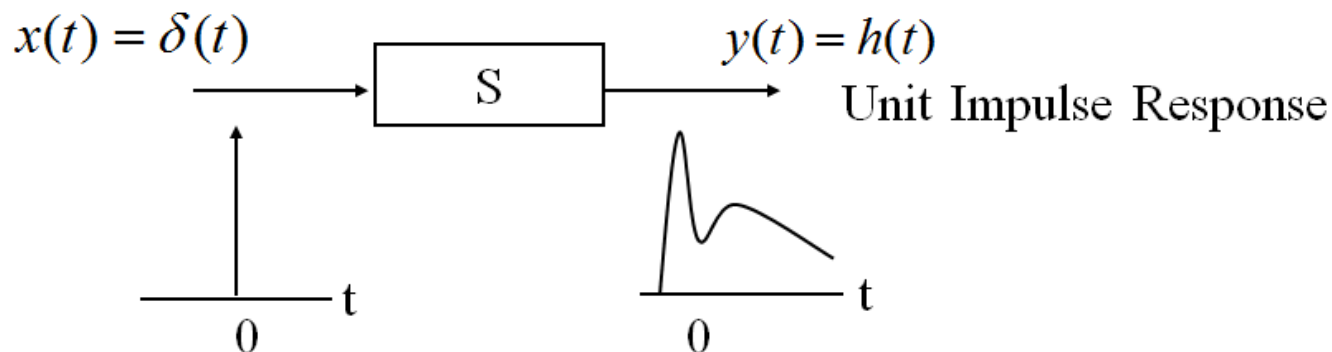
- **3.3**
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- **Tutorial problems**

- **Basic Problems with Answers 3.8**
- **Basic Problems 3.34**
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Review for Chapter 2

- Why to introduce **unit impulse response**?

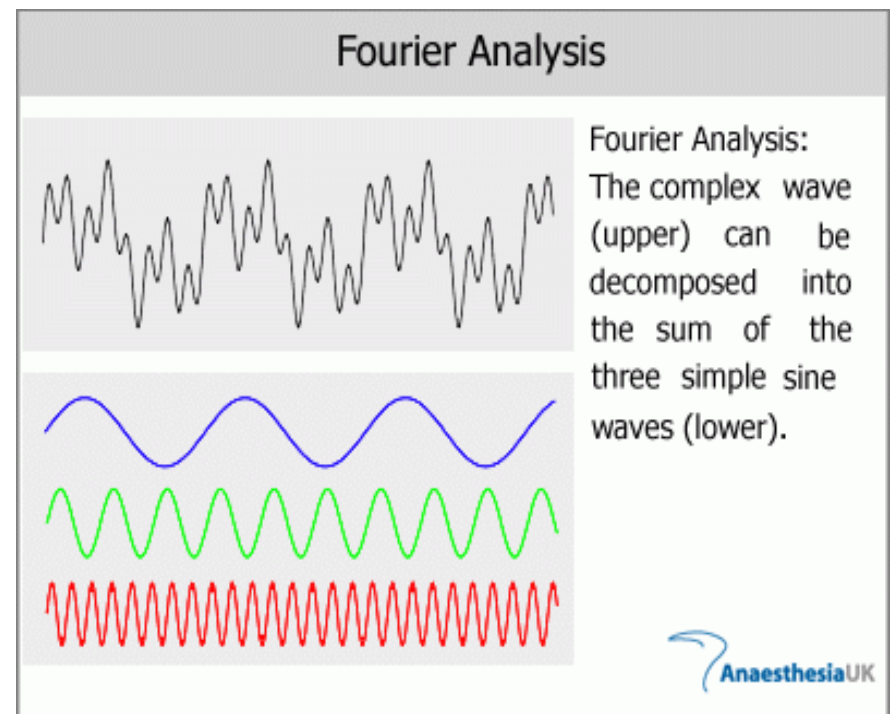


- Why to introduce convolution?
 - (DT or CT) Signal can be represented by a linear combination of **unit impulse response**
 - When it goes through the system, the output is computed via convolution of input signal and **unit impulse response**

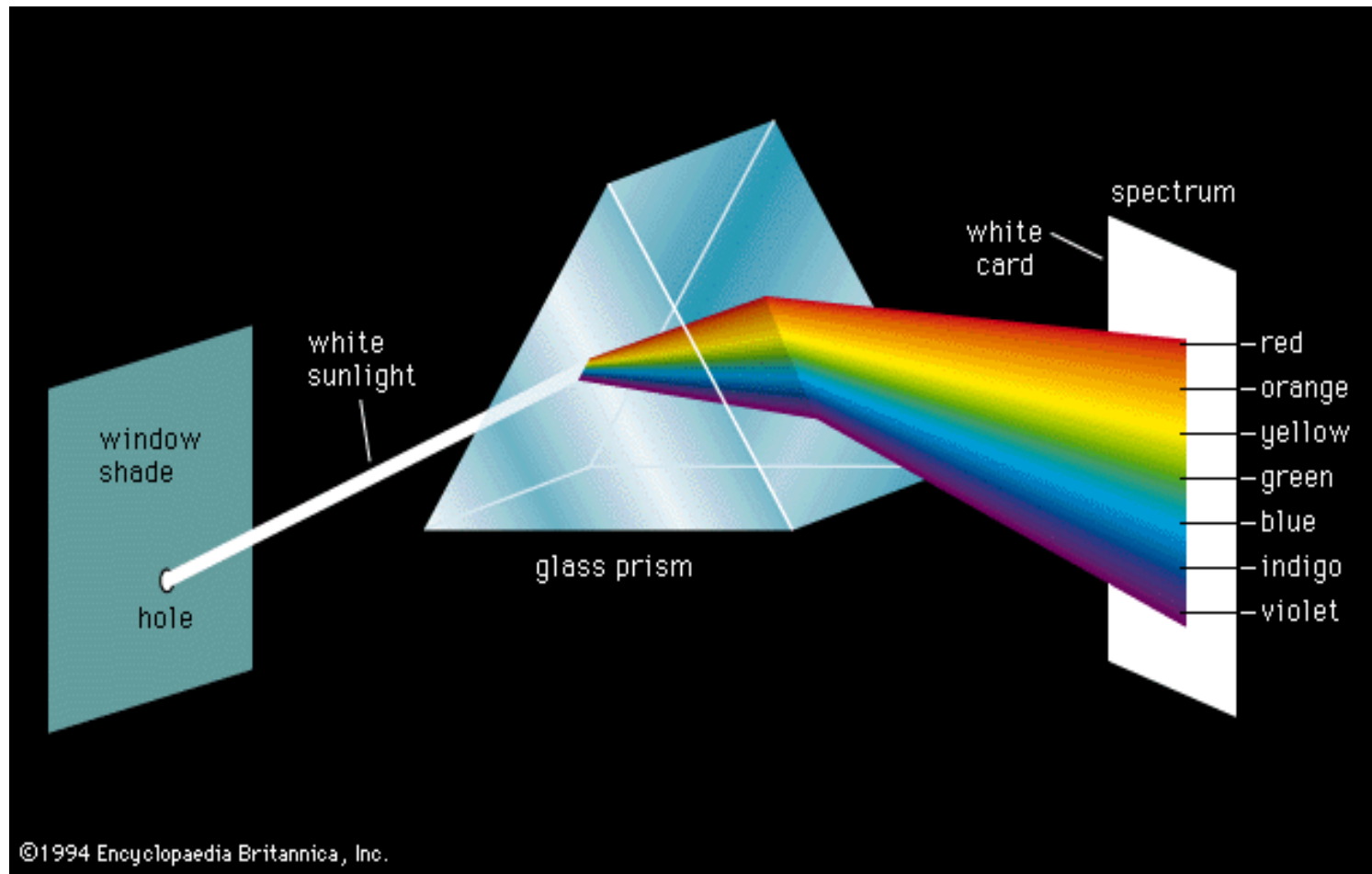
Chapter 3

Fourier Series Representation of Periodic Signals

Joseph Fourier

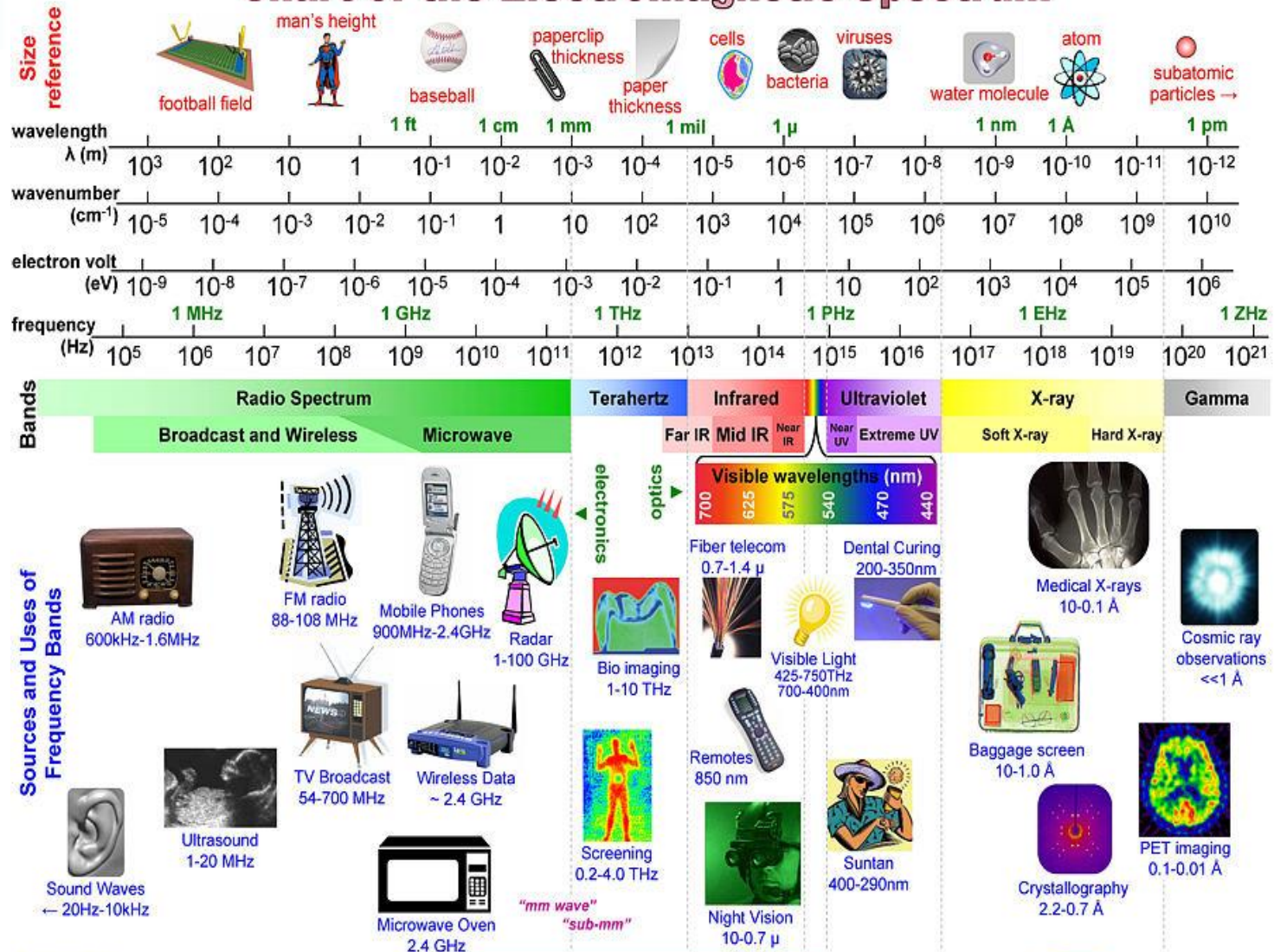


A Useful Analogy

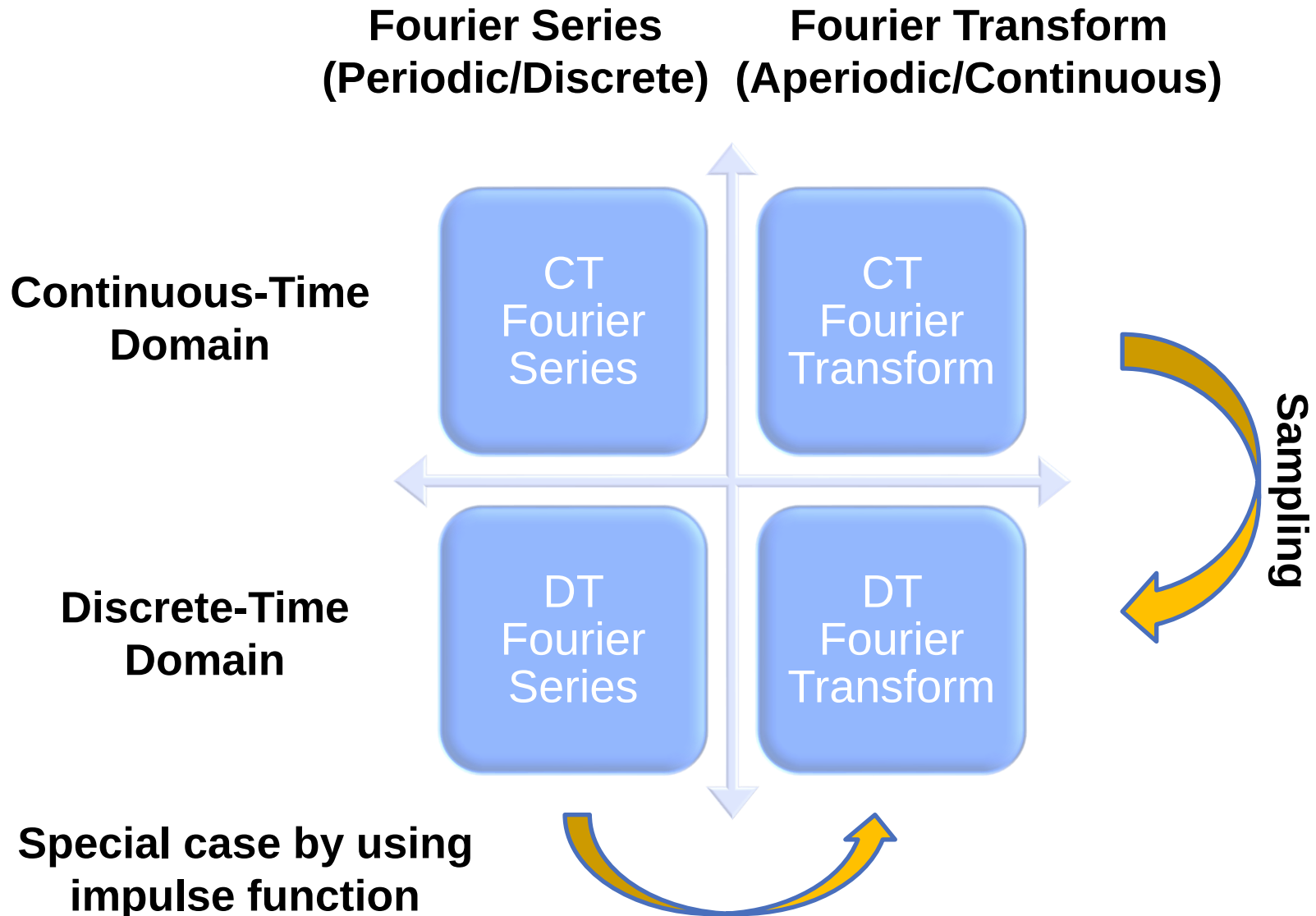


Example: Radio Spectrum

Chart of the Electromagnetic Spectrum



Overview on Frequency Analysis

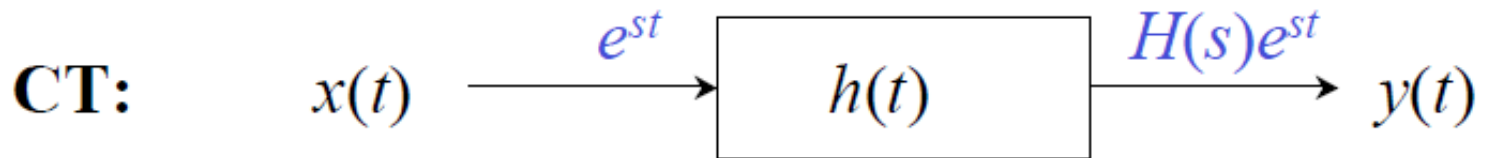


Two Special Signals for LTI Systems

$$\begin{array}{c}
 x(t) = e^{st} \longrightarrow \boxed{h(t)} \longrightarrow y(t) = \int_{-\infty}^{+\infty} h(\tau) e^{s(t-\tau)} d\tau \\
 = \left[\int_{-\infty}^{+\infty} h(\tau) e^{-s\tau} d\tau \right] e^{st} \\
 = \underbrace{H(s)}_{\text{eigenvalue}} \underbrace{e^{st}}_{\text{eigenfunction}}
 \end{array}$$

$$\begin{array}{c}
 x[n] = z^n \longrightarrow \boxed{h[n]} \longrightarrow y[n] = \sum_{m=-\infty}^{\infty} h[m] z^{n-m} \\
 = \left[\sum_{m=-\infty}^{\infty} h[m] z^{-m} \right] z^n \\
 = \underbrace{H(z)}_{\text{eigenvalue}} \underbrace{z^n}_{\text{eigenfunction}}
 \end{array}$$

System Functions $H(s)$ or $H(z)$



$$H(s) = \int_{-\infty}^{\infty} h(t)e^{-st} dt$$

$$x(t) = \sum a_k e^{s_k t} \longrightarrow y(t) = \sum H(s_k) a_k e^{s_k t}$$



$$H(z) = \sum_{n=-\infty}^{\infty} h[n]z^{-n}$$

$$x[n] = \sum a_k z_k^n \longrightarrow y[n] = \sum H(z_k) a_k z_k^n$$

Fourier and Beyond

Observation: if one signal can be written as the **linear combination** of e^{st} or z^n , we need **NOT** to calculate the convolution for the LTI output.

When $s = j\omega, z = e^{j\omega}$

$\Rightarrow e^{j\omega t}, e^{j\omega n}$: *Fourier Series*

When s or z is general complex number

$\Rightarrow$ *Laplace Transform & Z Transform*

A “Special” Class of Periodic Signals

CT Fourier Series: one periodic CT signal with **period T** can be written as

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

where $\omega_0 = 2\pi/T$.

$\{a_k\}$: Fourier series coefficients, which represent the strength of the component $e^{jk\omega_0 t}$.

$e^{jk\omega_0 t}$ is a signal with pure frequency $k\omega_0 \Rightarrow x(t)$ is a periodic signal with period T, it consists of components with different frequencies $k\omega_0$ and different weights.

Convergence of CT Fourier Series

What kind of periodic signals have Fourier series expansion?

Define $e(t) = x(t) - \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$,

- Fourier series expansion exists $\Leftrightarrow \exists \{a_k\}, e(t) = 0$ (D1)

Relaxation:

- Fourier series expansion exists $\Leftrightarrow \exists \{a_k\}, \int_T |e(t)|^2 dt = 0$ (D2)
- $D1 \subset D2$
- Two sufficient conditions for D2
 - $\int_T |x(t)|^2 dt < \infty$
 - Dirichlet condition

Dirichlet Conditions

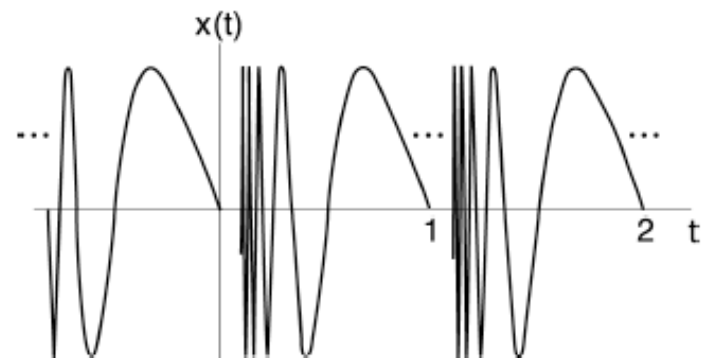
Condition 1. $x(t)$ is *absolutely integrable* over one period, i. e.

$$\int_T |x(t)| dt < \infty$$

Condition 2. In a finite time interval, $x(t)$ has a *finite* number of maxima and minima.

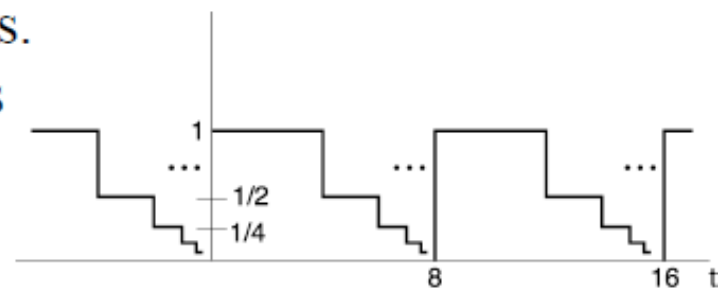
Ex. An example that violates Condition 2.

$$x(t) = \sin(2\pi/t) \quad 0 < t \leq 1$$



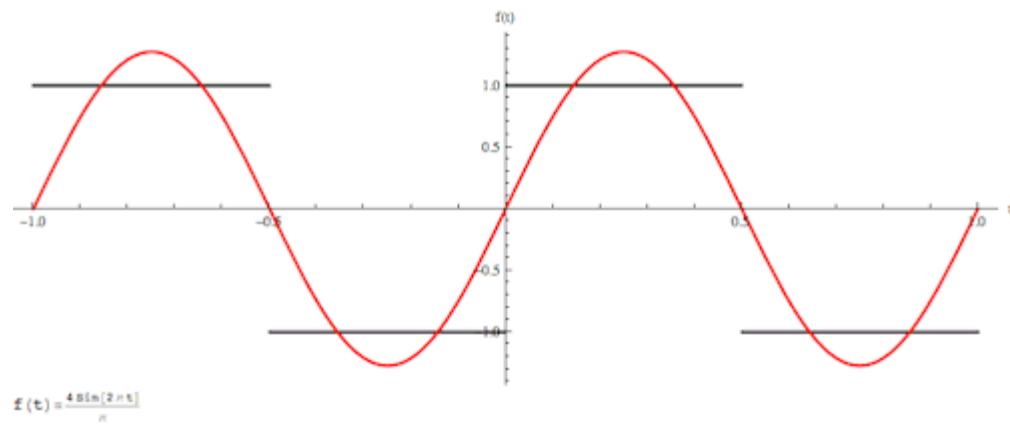
Condition 3. In a finite time interval, $x(t)$ has only a *finite* number of discontinuities.

Ex. An example that violates Condition 3.



**Almost all the periodic signals in practice
have Fourier series expansion!**





Question: How do we find the Fourier coefficients?

Let's first take a detour by studying a three-dimensional vector:

$$\vec{A} = A_x \hat{x} + A_y \hat{y} + A_z \hat{z},$$

$\hat{x}, \hat{y},$ and $\hat{z}$ are unit vectors.

How do we find the coefficients $A_x, A_y,$ and A_z ?

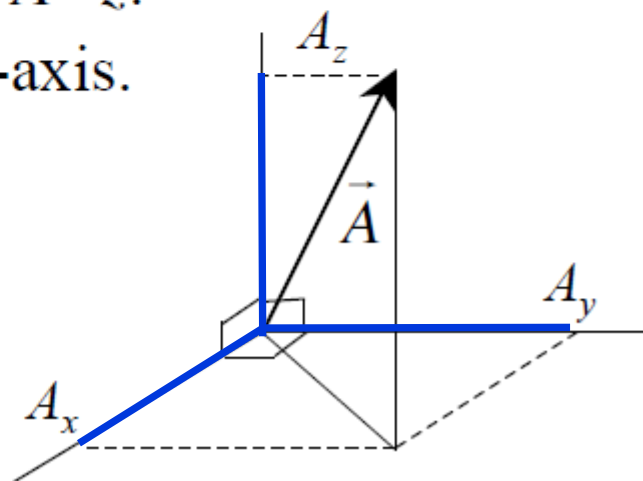
Easy, $A_x = \vec{A} \cdot \hat{x}, A_y = \vec{A} \cdot \hat{y},$ and $A_z = \vec{A} \cdot \hat{z}.$

Project the vector onto the x -, y -, and z -axis.

Why does it work this way?

Orthogonality:

$$\hat{y} \cdot \hat{x} = \hat{z} \cdot \hat{y} = \hat{x} \cdot \hat{z} \equiv 0$$



Inner Product of Exponential Signals

- Define inner product as

$$\langle e^{jk\omega_0 t} \cdot e^{jn\omega_0 t} \rangle = \frac{1}{T} \int_T e^{jk\omega_0 t} (e^{jn\omega_0 t})^* dt$$

- We have

$$\langle e^{jk\omega_0 t} \cdot e^{jn\omega_0 t} \rangle = 1 \quad (k = n)$$

$$\langle e^{jk\omega_0 t} \cdot e^{jn\omega_0 t} \rangle = 0 \quad (k \neq n)$$

- $\{e^{jk\omega_0 t} | \forall \text{ integer } k\}$ is the basis of periodic signal space with period T

How to Obtain Fourier Coefficients

- $x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} \Rightarrow \vec{A} = A_x \hat{x} + A_y \hat{y} + A_z \hat{z}$

- Notice

$$A_x = \vec{A} \cdot \hat{x}$$

- Similarly, we guess

$$a_k = \langle x(t) \cdot e^{jk\omega_0 t} \rangle$$

Let's double-check

$$\begin{aligned} & \langle x(t) \cdot e^{jk\omega_0 t} \rangle \\ &= \sum_{n=-\infty}^{+\infty} a_n \langle e^{jn\omega_0 t} \cdot e^{jk\omega_0 t} \rangle \\ &= a_k \end{aligned}$$



Important

CT Fourier Series Pair

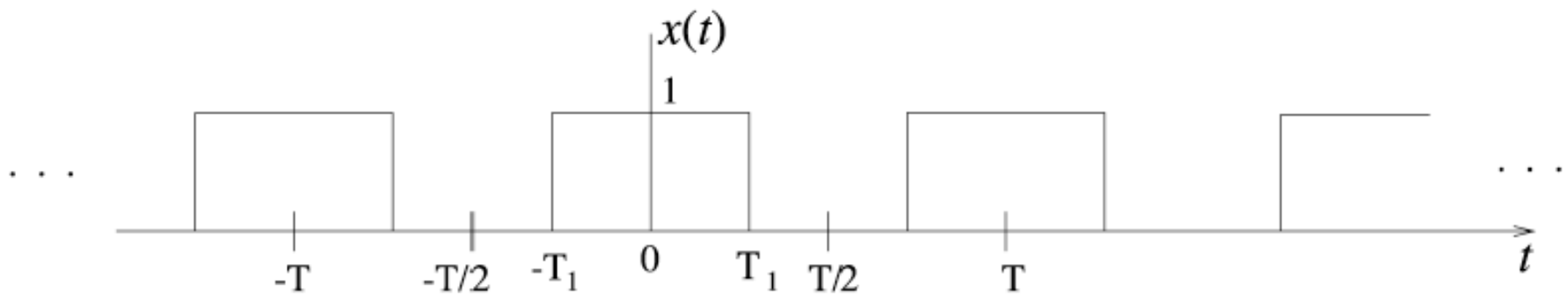
CT Fourier Series Pair

$$(\omega_o = 2\pi / T)$$

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_o t} \quad (\text{Synthesis equation})$$

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_o t} dt \quad (\text{Analysis equation})$$

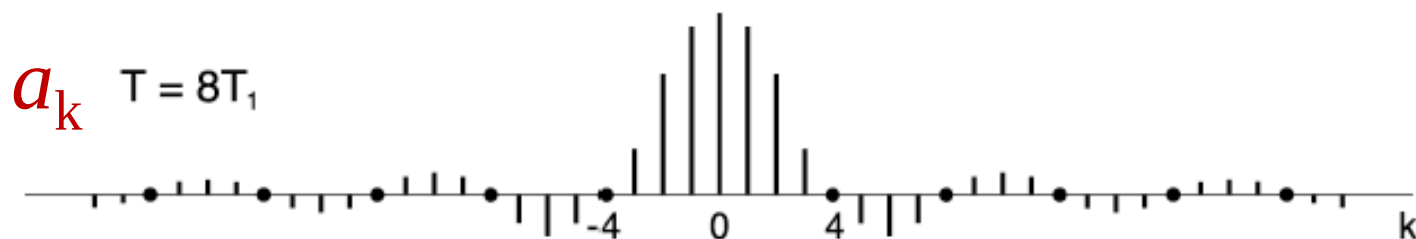
Example 3.5: Periodic Square Wave



$$a_o = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt = \frac{2T_1}{T}$$

$$k \neq 0 \quad a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-T_1}^{T_1} e^{-jk\omega_0 t} dt$$

$$(\omega_o = \frac{2\pi}{T}): \quad = -\frac{1}{jk\omega_o T} e^{-jk\omega_0 t} \Big|_{-T_1}^{T_1} = \frac{\sin(k\omega_o T_1)}{k\pi}$$



Example: Synthesis

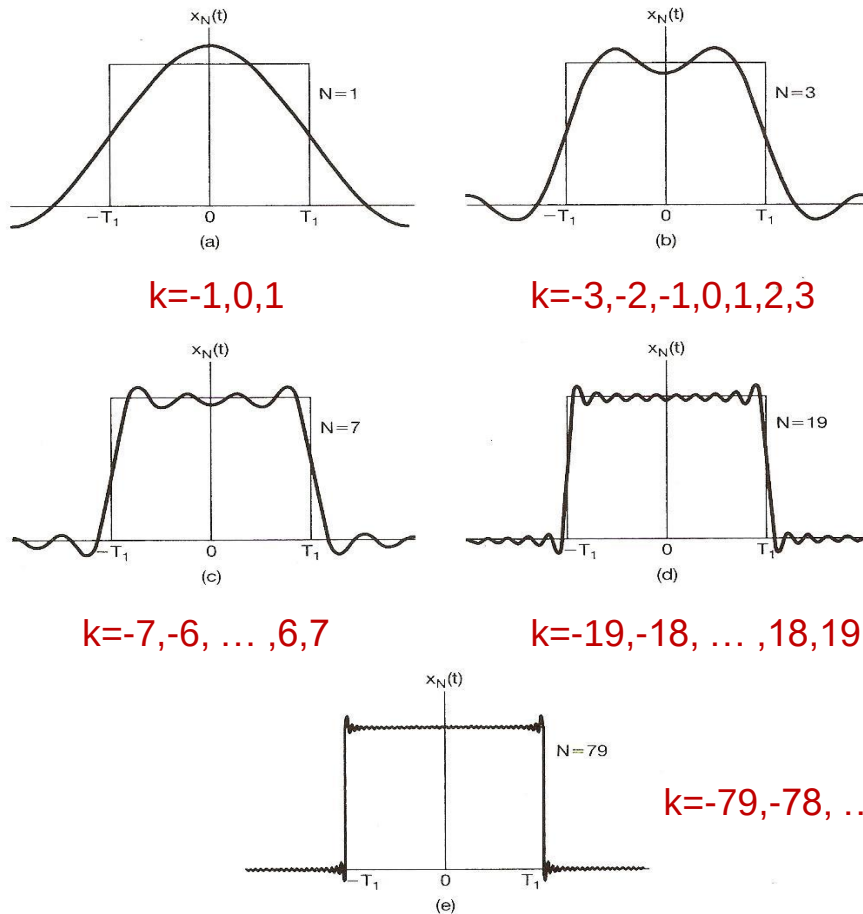
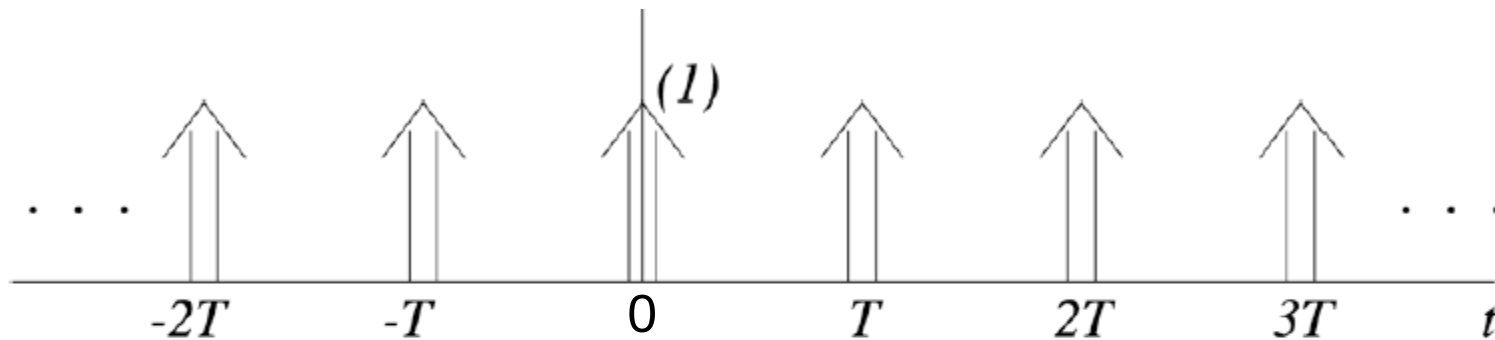


Figure 3.9 Convergence of the Fourier series representation of a square wave: an illustration of the Gibbs phenomenon. Here, we have depicted the finite series approximation $x_N(t) = \sum_{k=-N}^N a_k e^{jk\omega_0 t}$ for several values of N .

- **Gibbs Phenomenon:** the partial sum in the vicinity of the discontinuity exhibits ripples whose amplitude does not seem to decrease with increasing N

Periodic Impulse Train

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) \quad \text{— Sampling function, important for sampling later}$$



$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_0 t} dt$$

$$= \frac{1}{T} \quad \text{for all } k !$$

⇓

$$x(t) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} e^{jk\omega_0 t}$$

— All components have:

- (1) the same amplitude,
- &
- (2) the same initial phase.

Properties of Fourier Series

- Linearity

$x(t)$ and $y(t)$ are with the same period,

$$x(t) \leftrightarrow a_k \quad \text{and} \quad y(t) \leftrightarrow b_k,$$



$$\alpha x(t) + \beta y(t) \leftrightarrow \alpha a_k + \beta b_k$$

- Time shift (delay leads to linear phase shift)

$$x(t) \leftrightarrow a_k$$



$$x(t - t_0) \leftrightarrow b_k = a_k e^{-j\omega_0 t_0}$$

- Conjugate symmetry

$$x(t) \text{ is real and } x(t) \leftrightarrow a_k \Rightarrow a_{-k} = a_k^*$$

Proof:

$$a_{-k} = \frac{1}{T} \int_T x(t) e^{jk\omega_o t} dt = \left[\frac{1}{T} \int_T x^*(t) e^{-jk\omega_o t} dt \right]^* = a_k^*$$

$$\Downarrow a_k = \text{Re}\{a_k\} + j \text{Im}\{a_k\}$$

$$= |a_k| e^{j\angle a_k}$$

$\text{Re}\{a_{-k}\} + j \text{Im}\{a_{-k}\}$

$\text{Re}\{a_k\} - j \text{Im}\{a_k\}$

$\text{Re}\{a_k\}$ is even , $\text{Im}\{a_k\}$ is odd
 or $|a_k|$ is even, $\angle a_k$ is odd

- Time reversal

$$x(t) \leftrightarrow a_k \Rightarrow x(-t) \leftrightarrow b_k = a_{-k}$$

Observation: the effect of sign change for $x(t)$ and a_k are identical

- Time scaling

- α : positive real number

- $x(\alpha t)$: periodic with period T/α and fundamental frequency $\omega_0 \alpha$

$$x(t) = \sum_{-\infty}^{\infty} a_k e^{jk\omega_0 t} \Rightarrow x(\alpha t) = \sum_{-\infty}^{\infty} a_k e^{jk(\alpha\omega_0)t}$$

● Multiplication Property

$x(t)$ and $y(t)$ are with the same period,
 $x(t) \leftrightarrow a_k$ and $y(t) \leftrightarrow b_k$,



$$x(t) \cdot y(t) \leftrightarrow c_k = \sum_{l=-\infty}^{\infty} a_l b_{k-l} = a_k * b_k$$

Proof:

$$\begin{aligned} & \sum_{l=-\infty}^{+\infty} a_l e^{j l \omega_0 t} \cdot \sum_{m=-\infty}^{+\infty} b_m e^{j m \omega_0 t} \\ &= \sum_{l, m=-\infty}^{+\infty} a_l b_m e^{(l+m) \omega_0 t} \xrightarrow{l+m=k} \sum_{k=-\infty}^{+\infty} \left[\sum_{l=-\infty}^{+\infty} a_l b_{k-l} \right] e^{j k \omega_0 t} \end{aligned}$$

- Parseval Relation

$$\underbrace{\frac{1}{T} \int_T |x(t)|^2 dt}_{\text{Average Signal Power}} = \sum_{k=-\infty}^{\infty} \underbrace{|a_k|^2}_{\text{Power of component } e^{jk\omega_0 t}}$$

Observation: power is the same whether measured in the time-domain or the frequency-domain

Proof:

$$\begin{aligned} \frac{1}{T} \int_T |x(t)|^2 dt &= \left\langle \sum_{l=-\infty}^{+\infty} a_l e^{jl\omega_0 t}, \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} \right\rangle \\ &= \sum_{k=-\infty}^{+\infty} \left\langle a_k e^{jk\omega_0 t}, a_k e^{jk\omega_0 t} \right\rangle = \sum_{k=-\infty}^{+\infty} |a_k|^2 \end{aligned}$$

More Properties

- Frequency shifting

$$e^{jM\omega_0 t} x(t) \longleftrightarrow b_k = a_{k-M}$$

- Differentiation

$$\frac{dx(t)}{dt} \longleftrightarrow b_k = jk\omega_0 a_k$$

- Integration

$$\int_{-\infty}^t x(t) dt \longleftrightarrow b_k = \left(\frac{1}{jk\omega_0}\right) a_k$$

TABLE 3.1 PROPERTIES OF CONTINUOUS-TIME FOURIER SERIES

Property	Section	Periodic Signal	Fourier Series Coefficients
		$\left. \begin{array}{l} x(t) \\ y(t) \end{array} \right\} \begin{array}{l} \text{Periodic with period } T \text{ and} \\ \text{fundamental frequency } \omega_0 = 2\pi/T \end{array}$	$\begin{array}{l} a_k \\ b_k \end{array}$
Linearity	3.5.1	$Ax(t) + By(t)$	$Aa_k + Bb_k$
Time Shifting	3.5.2	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0} = a_k e^{-jk(2\pi/T)t_0}$
Frequency Shifting		$e^{jM\omega_0 t} = e^{jM(2\pi/T)t} x(t)$	a_{k-M}
Conjugation	3.5.6	$x^*(t)$	a_k^*
Time Reversal	3.5.3	$x(-t)$	a_{-k}
Time Scaling	3.5.4	$x(\alpha t), \alpha > 0$ (periodic with period T/α)	a_k
Periodic Convolution		$\int_{-\infty}^{\infty} x(\tau)y(t-\tau)d\tau$	$Ta_k b_k$
Multiplication	3.5.5	$x(t)y(t)$	$\sum_{l=-\infty}^{\infty} a_l b_{k-l}$
Differentiation		$\frac{dx(t)}{dt}$	$jk\omega_0 a_k = jk \frac{2\pi}{T} a_k$
Integration		$\int_{-\infty}^t x(\tau) d\tau$ (finite valued and periodic only if $a_0 = 0$)	$\left(\frac{1}{jk\omega_0}\right)a_k = \left(\frac{1}{jk \frac{2\pi}{T}}\right)a_k$
Conjugate Symmetry for Real Signals	3.5.6	$x(t)$ real	$\begin{cases} a_k = a_{-k}^* \\ \operatorname{Re}\{a_k\} = \operatorname{Re}\{a_{-k}\} \\ \operatorname{Im}\{a_k\} = -\operatorname{Im}\{a_{-k}\} \\ a_k = a_{-k} \\ \angle a_k = -\angle a_{-k} \end{cases}$
Real and Even Signals	3.5.6	$x(t)$ real and even	a_k real and even
Real and Odd Signals	3.5.6	$x(t)$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals		$\begin{cases} x_e(t) = \operatorname{Ev}\{x(t)\} & [x(t) \text{ real}] \\ x_o(t) = \operatorname{Od}\{x(t)\} & [x(t) \text{ real}] \end{cases}$	$\begin{array}{l} \operatorname{Re}\{a_k\} \\ j\operatorname{Im}\{a_k\} \end{array}$

Parseval's Relation for Periodic Signals

$$\frac{1}{T} \int_T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2$$